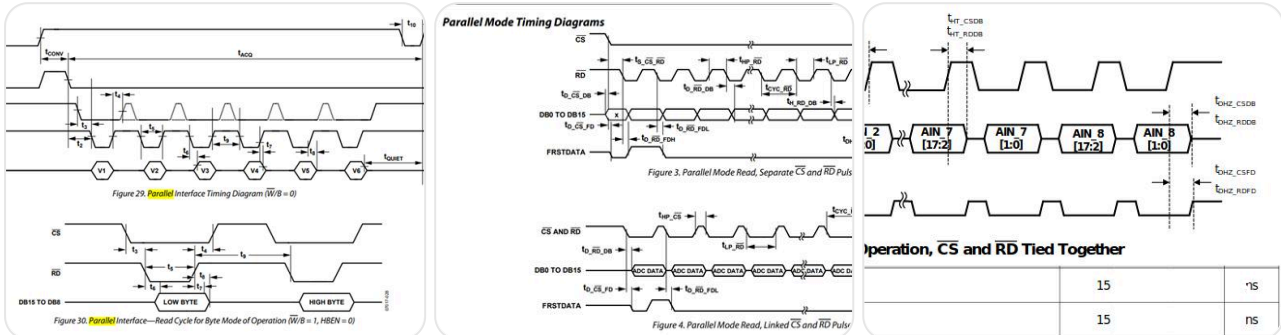


Great — now let's decode your **Figure 4, Figure 5, Figure 6** of ADS8588H in a **clear, step-by-step way**. I'll connect them so you understand the full flow 🔥



FIGURE 4 → Parallel Read (CS & RD Separate)



What this mode means

- **CS (Chip Select)** → Enables ADC
- **RD (Read)** → Tells ADC to output data



Both are **separate control signals**



Step-by-step flow



1. Conversion

- CONVST → start conversion
- BUSY → HIGH (ADC working)



2. Conversion complete

- BUSY → LOW
- Data ready inside ADC



3. Enable ADC

- CS → LOW
- ADC selected (but no data yet)



4. Read data

- RD → LOW
- Data appears on **DB[15:0]**

✓ After small delay (t_{D_Rddb})

5. Next channel

- RD pulse again (HIGH $\rightarrow$ LOW)

👉 Output changes:

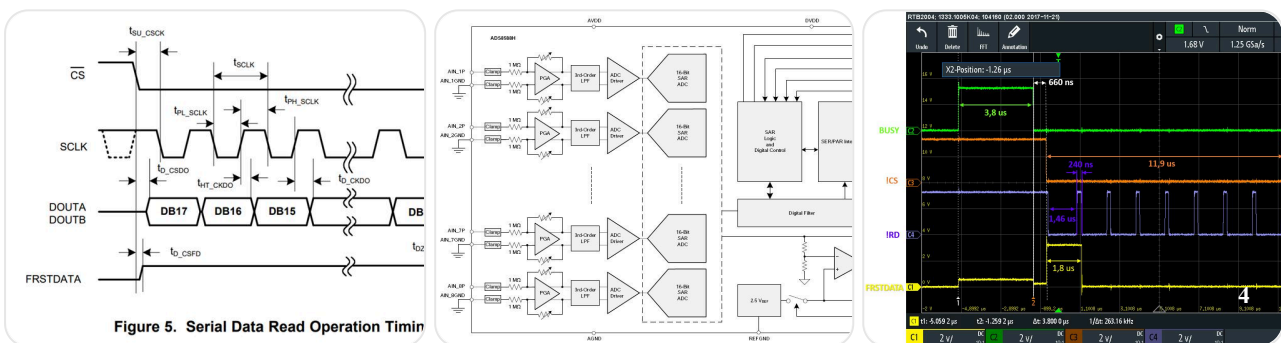
AIN1 $\rightarrow$ AIN2 $\rightarrow$ AIN3 $\rightarrow$...

Key idea

👉 CS = enable chip

👉 RD = fetch next data

FIGURE 5 $\rightarrow$ Serial Read Operation



What this mode means

- Data comes **bit-by-bit (serial)**
- Uses:
 - SCLK $\rightarrow$ clock
 - DOUTA / DOUTB $\rightarrow$ output lines

Step-by-step flow

1. Conversion

- Same as before
- BUSY HIGH $\rightarrow$ LOW

2. Start serial read

- CS → LOW
- 👉 Enables serial output

3. Clocking data

- Apply SCLK pulses
- 👉 Each clock → 1 bit comes out

4. Data sequence

- First 16 clocks → AIN1
- Next 16 clocks → AIN2
- continues...

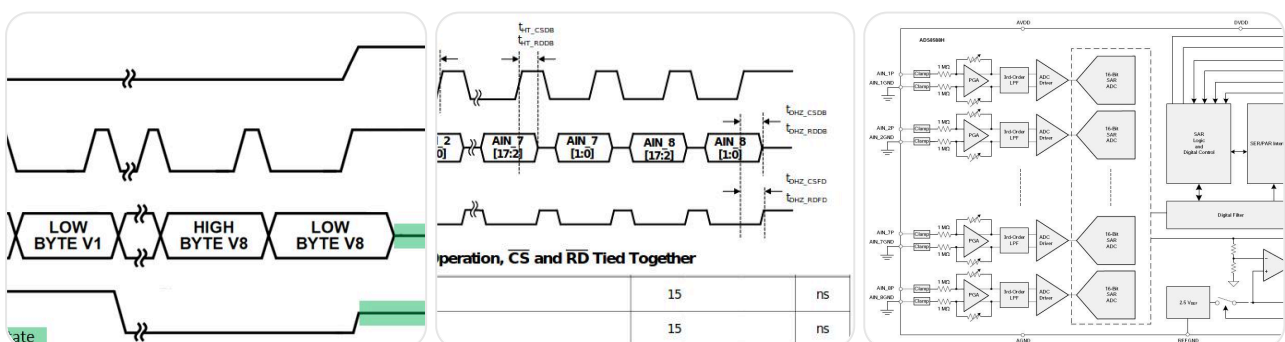
FRSTDATA signal

- 👉 Indicates:
 - “This is first channel (AIN1)”

Key idea

- 👉 Instead of RD pulses → use **clock pulses**

FIGURE 6 → Byte Mode Read



What this mode means

- Data is **16-bit**, but read as:
 - **High byte (8-bit)**
 - **Low byte (8-bit)**

Step-by-step flow

1. Conversion complete

- BUSY → LOW

2. First RD pulse

- RD LOW

 Output:

High Byte of AIN1

3. Second RD pulse

- RD LOW again

 Output:

Low Byte of AIN1

4. Next channel

- Next RD pulses:

AIN2 High → AIN2 Low → AIN3 High → ...

Key idea

 Each channel needs **2 RD pulses**

FINAL COMPARISON (VERY IMPORTANT)

Feature	Fig 4 (Parallel)	Fig 5 (Serial)	Fig 6 (Byte Mode)
Data format	16-bit full	Bit-by-bit	8-bit + 8-bit
Speed	Fastest	Slower	Medium
Control	CS + RD	CS + SCLK	CS + RD
Read method	1 RD = 1 channel	clocks	2 RD = 1 channel



MOST IMPORTANT TAKEAWAY



In ALL modes:

- BUSY LOW = data ready
 - Data comes in **fixed sequence** (AIN1 → AIN8)
 - You just **keep reading** (RD or clock)
-



If you want next level clarity:

I can:

- ✓ Draw **clean waveform** for AIN1 → AIN2 transition
- ✓ Explain **exact delays** like tSU, tH using your diagram arrows
- ✓ Or simulate step-by-step timing like your exam question

Just tell 👍